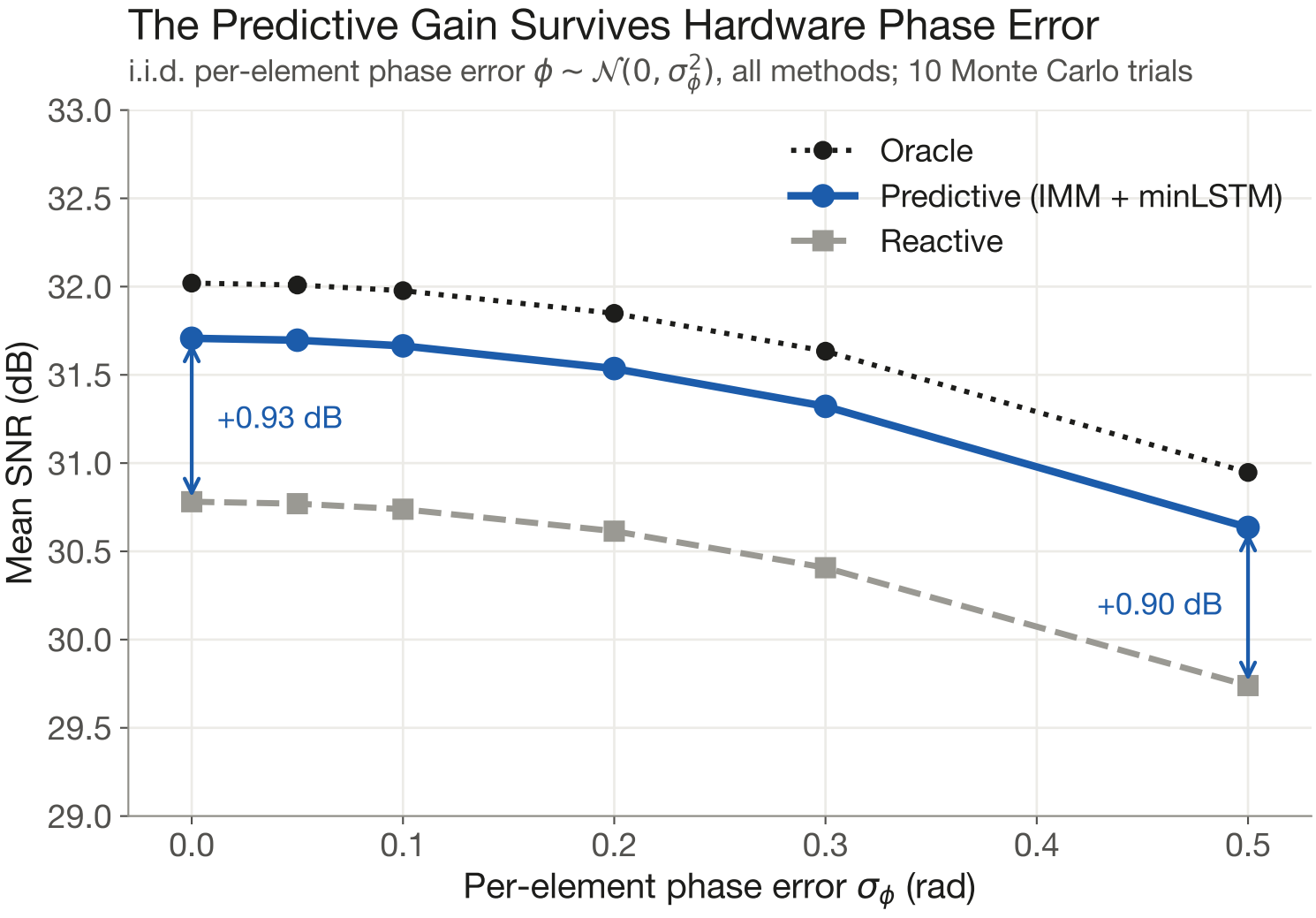
The Predictive Gain Survives Hardware Phase Error
i.i.d. per-element phase error $\phi \sim \mathcal{N}(0, \sigma_\phi^2)$, all methods; 10 Monte Carlo trials
Oracle
Predictive (IMM + minLSTM)
Reactive
+0.93 dB
+0.90 dB
Mean SNR (dB)
33.0
32.5
32.0
31.5
31.0
30.5
30.0
29.5
29.0
0.0
0.1
0.2
0.3
0.4
0.5
Per-element phase error σ_ϕ (rad)